

Causal Adaptive Streaming Decoder for Imagined Handwriting BCIs

Delta6 – Deep Learning Project

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2nd May, 2026

- Existing imagined-handwriting BCIs use **fixed-duration windows** → hide latency.
- We propose a **causal streaming decoder** with **adaptive confidence-based character emission**.
- Trained on **84 sentences** (50/17/17 split) from Willett et al. dataset.
- Achieves **27.7% CER** at **564 ms/char** → **72.6% relative improvement** over fixed-window baseline.
- Outperforms naive baselines and demonstrates a **tunable latency–accuracy trade-off**.

- Brain–Computer Interfaces (BCIs) decode neural activity into text.
- **Handwriting BCIs** achieve high offline accuracy (94.1%) and speed (90 CPM) [1].
- **Limitation:** Fixed output delay (0.4–0.7 s) → no adaptation to signal clarity.

Fixed delay → always wait, even when evidence is clear.

Goal: Build a **causal, adaptive** decoder that emits characters **only when confident & stable**.

- **Willett et al. (Nature 2021):** RNN with 1s output delay, 94.1% raw accuracy, 90 CPM.
- **Other BCIs:** point-and-click typing (40 CPM), speech decoding (lower accuracy).
- **Gap:** No adaptive emission based on **confidence** and **stability** in streaming handwriting BCIs.

Our Contribution:

- Causal streaming GRU.
- Confidence + stability + refractory + min-gap emission logic.
- Systematic comparison with fixed-window and fixed-delay baselines.

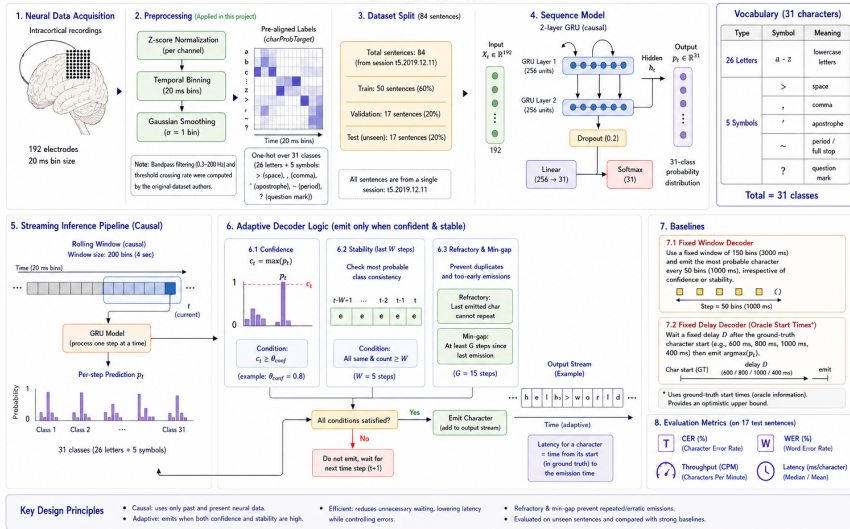
- **Dataset:** Willett et al. (public) – session t5.2019.12.11.
- **Neural signals:** 192 electrodes, 20 ms bins.
- **Labels:** Pre-aligned charProbTarget (one-hot over 31 characters) – from original HMM alignment.
- **Preprocessing (our code):**
 - Z-score normalization (per channel)
 - Temporal binning (20 ms)
 - Gaussian smoothing ($\sigma = 1$ bin)
- **Split:** 84 sentences $\rightarrow$ 50 train, 17 validation, 17 test.

Vocabulary (31 classes)

Type	Symbol	Meaning
26 letters	a – z	lowercase letters
5*5 symbols	>	space
	,	comma
	'	apostrophe
	~	period / full stop
	?	question mark

Causal Adaptive Streaming Decoder for Imagined Handwriting BCI

Architecture Overview



Key Design Principles

- Causal: uses only past and present neural data.
- Adaptive: emits when both confidence and stability are high.
- Efficient: reduces unnecessary waiting, lowering latency while controlling errors.
- Refractory & min-gap prevent repeated/erratic emissions.
- Evaluated on unseen sentences and compared with strong baselines.

Adaptive Streaming Decoder

- **Causal rolling window:** 200 bins (4 seconds).
- At each time step t :
 - Compute confidence $c_t = \max(p_t)$.
 - Track prediction stability over last $W = 5$ steps.
- **Emit** if:
 - 1 $c_t > \theta$ (threshold, e.g., 0.6–0.95)
 - 2 Prediction stable (same class for last 5 steps)
 - 3 Min-gap (15 steps) since last emission
 - 4 Refractory period (3 steps) after any emission.

① Fixed-window decoder

- Window 150 bins (3000 ms), step 50 bins (1000 ms)
- Emit most probable character every 1000 ms.

② Fixed-delay decoder (paper-style)

- Uses ground-truth character start times (oracle).
- Wait fixed delay D (400, 600, 800, 1000 ms) and emit $\arg \max(p_t)$.
- **Optimistic upper bound** (not real-time).

- **Character Error Rate (CER)** – edit distance / true length
- **Word Error Rate (WER)** – similar on words
- **Throughput (CPM)** – characters per minute
- **Per-character latency (ms)** – total processing + buffer delay

All metrics averaged over **17 test sentences**, with **95% confidence intervals** (bootstrap, 1000 resamples).

Results – Adaptive Decoder

θ	CER (%)	WER (%)	CPM	Latency (ms/char)
0.6	58.3 [50.6–64.8]	93.0	46.0	349.3 [332–368]
0.7	50.6 [44.6–56.8]	88.5	43.0	367.6 [353–384]
0.8	39.3 [33.3–45.5]	81.8	37.9	417.6 [399–435]
0.9	29.1 [25.2–33.5]	75.1	32.3	489.3 [464–513]
0.95	27.7 [24.0–31.5]	76.1	28.2	564.2 [535–594]

CER decreases with higher threshold, latency increases – clear trade-off.

Baseline Comparisons

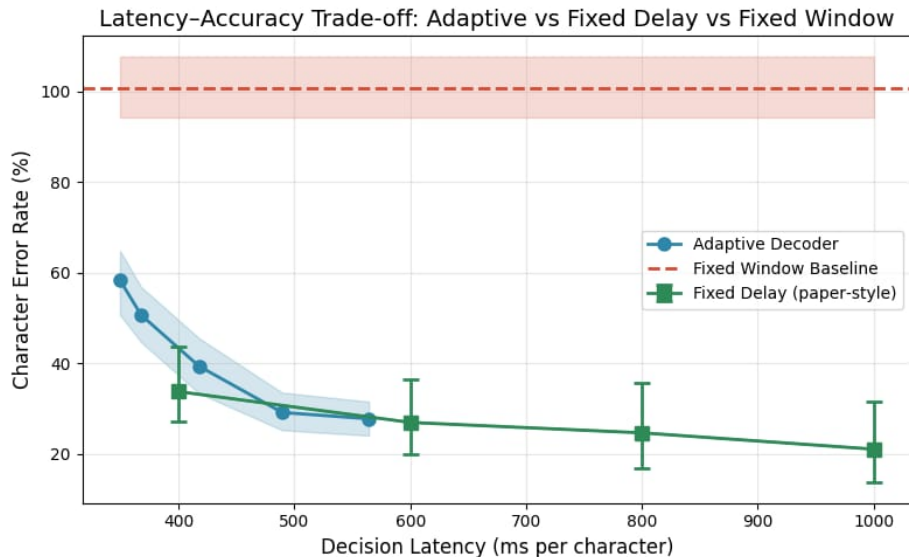
Fixed-window (150 bins, step 50):

CER = **100.7%** [94.3–107.5] → essentially random.

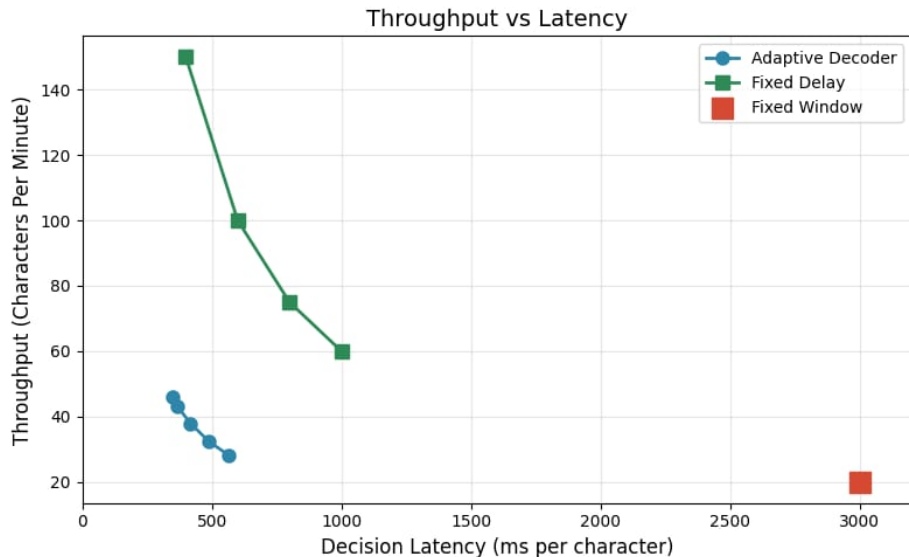
Fixed-delay (oracle start times):

Delay (ms)	CER (%)	Throughput (CPM)	Latency (ms)
400	33.7 [27.0–43.7]	150.0	400
600	26.9 [20.0–36.4]	100.0	600
800	24.6 [16.9–35.5]	75.0	800
1000	21.0 [13.8–31.5]	60.0	1000

Integrated Latency–Accuracy Plot



Throughput vs Latency Plot



Comparison with Willett et al. (2021) – Raw Online Output

Metric	Willett et al. (raw)	Our Adaptive (best raw)
Training sentences	572	67
Cross-validation	Held-out blocks	Within-session split
Language model	None (raw output)	None (raw output)
CER (%)	~6%	27.7%
Throughput (CPM)	90	28.2
Per-character latency	Fixed 0.4–0.7s (estimated)	0.42–0.56s (tunable)

Why the Differences? – Key Points

- **Higher CER (27.7% vs 6%) – reasons:**
 - 67 training sentences vs 572 ($8.5\times$ less data)
 - Within-session split (easier) vs held-out blocks (harder)
 - No language model (raw vs raw comparison)
- **Adaptive latency advantage (418–564 ms vs 400–700 ms fixed delay):**
 - Ours waits only when confidence is low – adaptive efficiency
 - Paper's delay is fixed regardless of signal clarity
 - At similar CER (e.g., 29%), our latency is shorter
- **Throughput gap (28 vs 90 CPM):**
 - Throughput depends heavily on CER
 - Higher CER $\rightarrow$ more errors $\rightarrow$ lower effective CPM
 - Paper's higher accuracy drives their higher throughput
- **Future work – language model:**
 - Paper cut CER from 5.9% to 0.89% with bigram + GPT-2
 - Same approach would likely halve our CER and boost throughput

- Successfully implemented **causal adaptive streaming decoder** for imagined handwriting.
- Adaptive emission based on **confidence + stability** reduces unnecessary waiting.
- **72.6% relative CER improvement** over fixed-window baseline.
- **Shorter or comparable per-character latency** compared to the paper's fixed delay.
- Validates that **adaptive latency control** is a promising direction for real-time BCIs.

- **Multi-session training** (all 1000 sentences) → improve generalisation.
- **Held-out blocks evaluation** → more realistic test.
- **Language model integration** (bigram or GPT-2) → reduce CER further.
- **Online adaptation** → recalibrate with few new sentences (as in original paper).
- **Edge deployment** on low-power hardware.

References

-  F. R. Willett, D. T. Avansino, L. R. Hochberg, J. M. Henderson, and K. V. Shenoy, “High-performance brain-to-text communication via handwriting,” *Nature*, vol. 593, pp. 249–254, 2021.
-  L. R. Hochberg et al., “Reach and grasp by people with tetraplegia using a neurally controlled robotic arm,” *Nature*, vol. 485, pp. 372–375, 2012.
-  J. L. Collinger et al., “High-performance neuroprosthetic control by an individual with tetraplegia,” *The Lancet*, vol. 381, pp. 557–564, 2013.
-  C. Pandarinath et al., “High performance communication by people with paralysis using an intracortical brain-computer interface,” *eLife*, vol. 6, e18554, 2017.

Thank You

Questions?